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Roll No. : .....

ACC No.

337551(37)

B. E. (Fifth Semester) Examination, April-May, 2016

(New Scheme)

(Mech. Engg. Branch)

MACHINE DESIGN-I

Time Allowed : Four hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) of each question is compulsory. For remaining as directed.*

1. (a) Define notch sensitivity. 2

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(Attempt any two parts from Part (b), (c) & (d))

(b) A 25 mm dia shaft is made of forged steel 30 C8, $S_{ut} = 600$ N/mm^2 there is a step in the shaft and the theoretical SCF at the step is 2·1. The notch sensitivity factor is 0·84. Determine the endurance limit of the shaft if it is subjected to a reversed bending moment. 7

(c) Write short notes on : 7

(i) Endorance diagram

(ii) Maximum shear stress theory

(iii) Distortion Energy Theory

(d) A machine component is subjected to fluctuating stress that varies flux 40 to 100 N/mm^2. The corrected endurance limit stress for the machine component is 270 N/mm^2. The ultimate tensile strength and yield strength of the material are 600 and 450 N/mm^2. Find the factor of safety using :

(i) Geiber theory

(ii) Soderberg line

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(iii) Goodman line

Also find factor of safety against static failure. 7

(a) Write different types of keys. 2

(Attempt any **two** parts from Part (b), (c) & (d))

(b) Design a cotter joint for carrying an axial load of 10 kN. 7

(c) Design a CI protective flange coupling to connect two shafts in order to transmit 7·5 kW at 720 rpm. The permissible stresses are :

Permissible shear stress for shaft, bolt and key material = 33 MPa, permissible crushing stress for bolt and key material = 60 MPa, permissible shear stress for CI = 15 MPa. 7

(d) Design a knuckle joint to connect two tension rods subjected to an axial load of 15 kN. 7

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3. (a) What is a cone clutch? 2

(Attempt all parts)

(b) Differentiate between shaft and axle. 4

(c) A mild shaft transmit 23 kW at 200 rpm. It carries a central load of 900 N and is simply supported between the bearings 25 m apart. Determine the size of the shaft, if the allowable shear stress is 42 MPa and the max. tensile or compressive stress is not to exceed 56 MPa. What is the size of shaft if it subjected to gradually applied load? 10

Or

Design a single plate clutch both side active which is used to transmit 10 kW at 4500 rpm in a truck.

4. (a) What is a power screw? 2

(Attempt all parts)

(b) What are different forms of screw threads? 4

(c) The lead screw of a lathe has trapezoidal threads. To drive the total carriage the screw has to exert an

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axial force of 20 kN. The thrust is carried by the collar. The length of the lead screw is 1·5 m. Coefficient of friction at the collar and nut are 0.1 and 0·15 respectively. Suggest suitable size of screw and height of the nut if the permissible bearing pressure is 4 MPa. Check for max. shear stress. 10

Or

A wall bracket is attached to the wall by means of four bolts as shown in fig. two at *A* and two at *B*. Assuming that the bracket is held against the wall and prevented from tipping about C by all four bolts and using allowable tensile stress in bolts 35 N/mm². Determine the size of bolts on the basis of max. principal stress theory.

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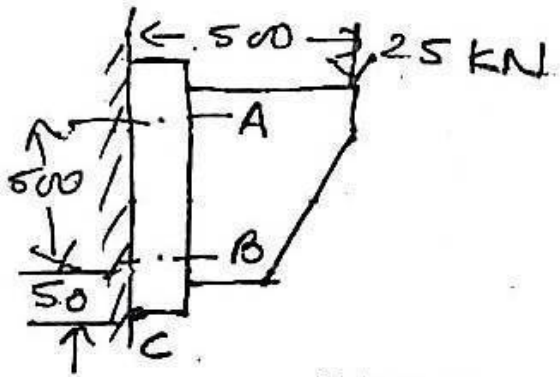

5. (a) Write different types of riveted joint. 2

(Attempt all parts)

(b) Derive an expression for a welded joint subjected to an eccentric loading to plane of weld. 4

(c) Find the effficiency of the following riveted joint :

(i) Single riveted lap joint of 6 mm plate with 20 mm diameter rivet having pitch of 25 mm.

(ii) Double riveted lap joint of 6 mm plate with 20 mm dia rivet having a pitch of 6 mm. Tensile stress of plate 120 MPa, shear stress 90 MPa and crushing stress 180 MPa. 10

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Or

Determine the length of the weld run for a plate of size 120 mm wide and 15 mm thick tobe welded to another plate by means of :

(i) A single transverse weld

(ii) Double parallel fillet welds when the joint is subjected to variable load.

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